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RESEARCH PAPER

Short-term aerobic pretreatment of swine manure enhances short-chain fatty acid production in anaerobic fermentation: performance and mechanisms

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Abstract

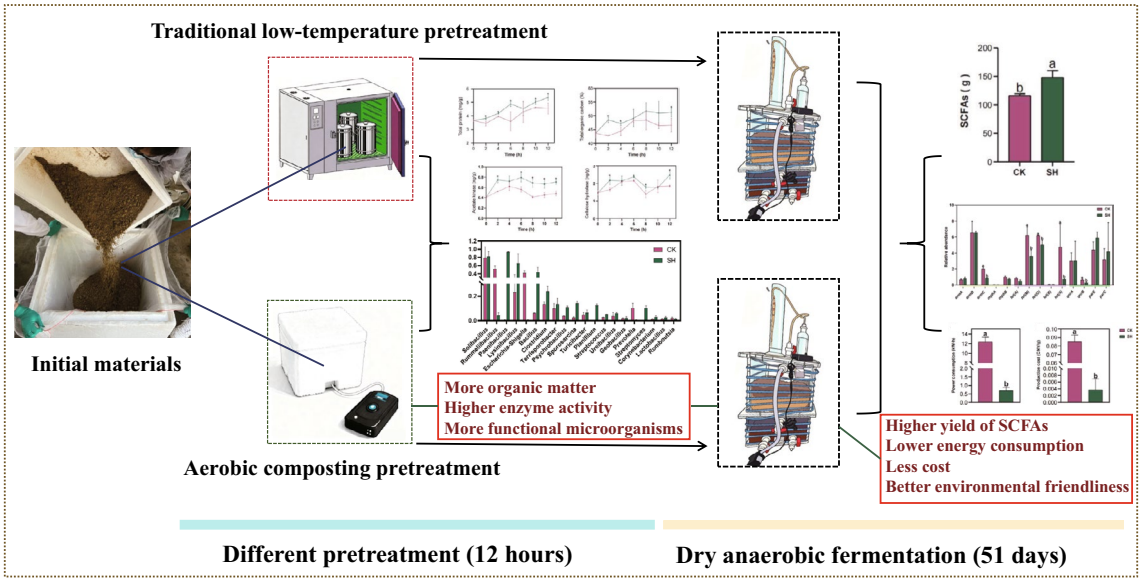
Facing the dual challenges of swine manure pollution control and carbon resource recovery under intensive farming conditions, conventional anaerobic fermentation technologies are limited in their application for high-value conversion due to low hydrolysis efficiency and high operational costs. To address these limitations, this study proposed a self-heating strategy using short-term aerobic pretreatment, aiming to optimize substrate pretreatment and enhance the synthesis of short-chain fatty acids during subsequent acidogenic fermentation. The results demonstrated that this pretreatment approach rapidly activates aerobic microbial metabolism and significantly improves the substrate's overall properties, as reflected by a decreased pH and increased conductivity, moisture content, total protein, and total organic carbon, and effectively promotes the solubilization and hydrolysis of organic matter, thereby increasing the bioavailability of substrates for anaerobic fermentation. Microbial community analysis further revealed that the pretreatment process selectively enriched hydrolytic-acidogenic bacteria, including *Clostridium*, *Bacillus*, *Paenibacillus*, and *Streptomyces*, while synergistically enhancing the activities of key metabolic enzymes such as cellulase, protease, and acetate kinase. These effects systematically strengthened the metabolic pathways from substrate degradation to acid synthesis. As a result, the short-term aerobic pretreatment achieved a significant improvement in SCFAs production and energy efficiency. Specifically, the total SCFAs yield increased by 21.31%±4.23%, while energy consumption declined by 94.48%±1.48%. In addition, the unit acid production cost amounted to merely 4.37%±0.15% of that required for the conventional low-temperature thermal treatment. Moreover, it significantly reduced the abundance of high-risk antibiotic resistance genes in fermentation residues, highlighting its environmental benefits. This study elucidates the synergistic mechanism of short-term aerobic pretreatment across "substrate–microorganism–enzyme" dimensions, confirming it as an efficient, economical, and sustainable pretreatment technology for swine manure resource recovery.

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Graphical abstract



Keywords Swine manure · Aerobic pretreatment · Anaerobic fermentation · Short-chain fatty acids

Introduction

Intensive and large-scale livestock production has led to a sharp increase in the generation of livestock manure, which has become a major source of agricultural non-point pollution [18, 33]. As one of the primary pollutants, swine manure is characterized by its large output, high organic matter concentration, and significant pathogen content [20, 28]. If not properly managed, it can pose serious risks to water, soil, and atmospheric environments [8, 36]. Current treatment methods include direct land application, aerobic composting, and anaerobic digestion [16, 49, 52]. However, direct application carries risks such as nutrient loss, pathogen transmission, and weed seed dispersal [27, 47]. Aerobic composting can produce organic fertilizer, but it is energy-intensive and results in substantial carbon loss in the form of CO_2, leading to low resource efficiency and demonstrating that this increases the carbon footprint [54]. Therefore, exploring technical pathways that not only treat swine manure effectively but also enable efficient carbon recovery and energy utilization is of great practical significance for promoting cleaner production and sustainable development in the livestock sector.

Anaerobic digestion, a mature technology, has long been used for energy recovery from livestock manure by converting organic matter into biogas (primarily methane) through microbial communities under anaerobic conditions [19, 30]. This process reduces pollutants, while the resulting digestate can serve as a quality fertilizer, facilitating nutrient recycling [53]. However, conventional methane-producing anaerobic digestion faces several limitations, including long hydraulic retention times (typically 20–40 days), large reactor volume requirements, slow system startup, and sensitivity to feedstock properties such as particle size and biodegradability [24]. Particularly for lignocellulose-rich swine manure [41], the slow biodegradation rate limits methane yield and organic loading capacity, constraining further improvements in treatment efficiency and economic viability.

To address these limitations, research focus has shifted toward the intermediate stage of anaerobic digestion—acidogenic fermentation, which aims to selectively convert complex organic matter into high-value short-chain fatty acids (SCFAs) such as acetate, propionate, and butyrate [40, 50]. SCFAs are not only important feedstocks for the chemical and pharmaceutical industries but also serve as high-quality carbon sources for biological nitrogen and phosphorus removal, offering significantly higher market value and broader application prospects compared to biogas [35, 45]. By optimizing the hydrolysis-acidogenesis stage and suppressing methanogenic activity, high SCFAs accumulation can be achieved [14]. Nevertheless, acidogenic fermentation still faces two major bottlenecks in practical applications: technically, the recalcitrant lignocellulosic structure of swine manure limits hydrolysis efficiency [41], resulting in low SCFAs yields; economically, conventional pretreatment methods—such as thermal, microwave, chemical, or enzymatic treatments—often involve high energy

consumption, chemical inputs, and costs, or may generate inhibitory by-products, thereby hindering engineering feasibility [42, 50].

Thus, there is a critical need to develop an economically viable and environmentally friendly pretreatment method to overcome the technical and economic barriers in acidogenic fermentation of swine manure. This study proposes and aims to validate short-term aerobic pretreatment as a cost-effective approach. The core concept is to utilize the bioheat generated from the short-term, limited-aeration aerobic degradation of organic matter in swine manure to achieve mild heating, activate hydrolytic and acidogenic microorganisms, and partially disrupt the manure structure, thereby reducing reliance on external energy and chemical additives. Based on this rationale, the main objectives of this study are: (1) to evaluate the effect of short-term aerobic pretreatment on SCFAs production during subsequent anaerobic fermentation; (2) to elucidate the underlying mechanisms by examining changes in the physicochemical properties and enzyme activities of the pretreated substrate; (3) to reveal the microbiological mechanisms via analysis of microbial community structure; and (4) to assess the environmental and economic advantages of this method compared to conventional pretreatments. Through systematic investigation of these aspects, this study aims to provide an innovative and potentially applicable technical pathway for the high-value resource recovery of swine manure.

Materials and methods

Experimental design and sample collection

In this study, two pretreatment methods—low-temperature thermal treatment (70 °C in an oven) and short-term aerobic pretreatment—were applied to the fermentation substrate, with experimental groups detailed in Table 1. Throughout the pretreatment process, physicochemical properties and microbial community dynamics of the substrate were continuously monitored. The experimental procedure, as illustrated in Figure A.1, was conducted as follows:

Fresh swine manure and wheat straw powder were mixed at a 5:1 ratio (w/w, wet basis) to form the fermentation substrate. A total of 30 kg of the substrate was evenly divided into three portions, each placed into a 10 L stainless steel container and subjected to low-temperature thermal treatment at 70 °C in an oven, designated as the control group (CK). Sampling was performed at 0, 2, 4, 6, 8, 10, and 12 h after the start of treatment. For each sampling event, 20 g of substrate was collected from each container, with 15 g stored at –20 °C and 5 g at –80 °C for subsequent analysis. Simultaneously, another 30 kg of the same substrate was placed in a foam incubation box equipped with bottom aeration for short-term aerobic pretreatment (for detailed descriptions, please refer to the Supplementary Materials Text A.1), designated as the SH group. When the core temperature reached 70 °C (after 22 h), this time point was recorded as 0 h. Sampling was then conducted at 0, 2, 4, 6, 8, 10, and 12 h using a five-point sampling method. A total of 60 g of sample was collected from the core area, homogenized, and divided into three replicates, with 15 g stored at –20 °C and 5 g at –80 °C. During the pretreatment phase, a socket capable of accurately measuring electrical power loss (BULL, Y201G, China) was used to monitor and record the energy consumption of the two treatment modes.

Table 1 Group information

Group	Initial materials	The method of preprocessing	The method of acid-producing fermentation
CK	Fresh swine manure + wheat straw powder	Low-temperature heating pretreatment	Dry anaerobic fermentation
SH	Fresh swine manure + wheat straw powder	Short-term aerobic pretreatment	Dry anaerobic fermentation

After pretreatment, 5 kg of substrate from each pretreatment group was loaded into a dry acidogenic fermentation reactor (for detailed descriptions, please refer to the Supplementary Materials Figure A.2). The reactor was sealed and then purged with high-purity nitrogen at a volume equivalent to several times that of the reactor to displace the residual atmosphere; the sealed condition was maintained for 24 h to ensure complete removal of residual oxygen. Thereafter, 7 L of distilled water was added to the reactor to serve as the circulating leaching solution. Each experimental group included three biological replicates, and the fermentation lasted 51 days. Based on leachate replacement time points (days 9, 21, 43, and 51), the fermentation process was divided into four phases: Phase I (days 1–9), Phase II (days 10–21), Phase III (days 22–43), and Phase IV (days 44–51). Daily, 20 mL of leachate was collected from the bottom sampling valve of each reactor for SCFAs analysis (After each sample collection, the system was replenished with distilled water). The samples were allocated as follows: 10 mL stored at 4 °C and was used for enzyme activity determination, 5 mL at –20 °C, and 5 mL at –80 °C. On day 51, reactor lids were opened, and 50 g of solid residue was collected from each reactor using the five-point sampling method and stored at –80 °C for DNA extraction and metagenomic sequencing.

Determination of short-chain fatty acids

The determination of SCFA was carried out by gas chromatography [37], which was equipped with a hydrogen flame

ionization detector (FID) (Intuvo 9000, Agilent, USA). The analytical procedure was as follows: A 1.5 mL aliquot of liquid sample was vortexed and allowed to stand for 30 min, followed by centrifugation at 10,000 × g for 15 min. Then, 1 mL (V_1) of the supernatant was mixed with 0.2 mL (V_2) of 25% metaphosphoric acid and 23.3 µL (V_3) of 210 mmol/L crotonic acid (as internal standard). After vortex mixing, the mixture was incubated at 4 °C for 30 min and centrifuged at 8,000 × g for 10 min. Subsequently, 0.3 mL (V_4) of the supernatant was combined with 0.9 mL (V_5) of chromatographic-grade methanol, vortexed, and centrifuged at 8,000 × g for 5 min. The resulting supernatant was filtered through a 0.22 µm nylon syringe filter into a 1.5 mL EP tube for GC analysis. SCFAs were identified based on retention time and quantified using peak areas against a standard curve. Liquid samples were collected daily from each reactor, and VFA concentrations were determined (expressed as g COD/L VS). Cumulative VFA production ($CumVFA_t$) was calculated using the following mass balance equation, which corrects for the VFA mass removed during periodic sampling: $CumVFAt = CumVFA_{t-1} + (C_t \times V_t) - (C_{t-1} \times V_{sample})$, where C_t and C_{t-1} are the measured VFA concentrations (g COD/L VS) at time "t" and "t-1", respectively; V_t is the liquid volume (L) in the reactor at time "t" after sampling; and V_{sample} is the volume of sample withdrawn at the previous sampling time (t-1). The initial cumulative production ($CumVFA_0$) was set to zero. This calculation accurately reflects the total VFA generated over the experimental period, rather than merely the residual concentration remaining in the reactor. Six SCFAs were analyzed in this study: acetate, propionate, isobutyrate, butyrate, isovalerate, and valerate. The GC conditions were set as follows: injector temperature 220 °C, injection volume 1 µL, split ratio 50:1; carrier gas: high-purity argon, column flow rate 1 mL/min; FID temperature 250 °C, hydrogen flow 40 mL/min, air flow 400 mL/min, makeup gas (argon) flow 35 mL/min. The separation was performed on an HP-FAPP capillary column (30 m × 0.53 mm × 1.0 µm) with the following temperature program: initial temperature 100 °C, increased at 20 °C/min to 190 °C, held for 1.5 min, yielding a total run time of 6 min.

Determination of physicochemical properties of the fermentation substrate after pretreatment

The pH and electrical conductivity (EC) of the fermentation substrate samples were determined using potentiometric methods [48]. Specifically, pH was measured in a 1:5 (w/v) aqueous extract using a calibrated pH meter, while EC was measured on the same extract using a conductivity meter to evaluate soluble salt content. Moisture content was determined by the constant weight method, where fresh samples were dried in an oven at 105 °C until constant weight was achieved. Total organic carbon (TOC) was quantified by the potassium dichromate oxidation method [4] with external heating, in which organic carbon was oxidized and the residual dichromate was titrated. Total protein content was determined using the Kjeldahl method [39].

Determination of enzyme concentration of fermentation substrates after pretreatment

The concentrations of four key metabolic enzymes (α-glucosidase, protease, cellulase and acetate kinase) were quantified using a double-antibody sandwich enzyme-linked immunosorbent assay (ELISA). Approximately 0.1 g of solid sample was accurately weighed and mixed with an appropriate amount of pre-cooled extraction buffer. The mixture was subjected to ultrasonic cell disruption (300 W, total duration 5 min, using a cycle of 10 s operation and 5 s pause) at 4 °C. The resulting homogenate was centrifuged at 10,000 × g for 10 min at 4 °C, and the supernatant was collected as the test sample. Then, 50 µL of the test sample or standard was precisely added to the microplate wells pre-coated with specific antibodies. After sealing with a plate seal, the plate was incubated at 37 °C for 30 min. Following incubation, the liquid in the wells was discarded, and each well was washed with 300 µL of washing buffer, allowed to stand for 30 s, and then emptied. This washing procedure was repeated five times. After pat-drying, 50 µL of horseradish peroxidase-labelled detection antibody was added to each well except the blank wells. The plate was resealed and incubated at 37 °C for another 30 min. After five additional washes and pat-drying, 50 µL of hydrogen peroxide substrate solution and 50 µL of 3,3′,5,5′-tetramethylbenzidine (TMB) chromogenic substrate were sequentially added to each well. The plate was gently shaken to mix and then incubated at 37 °C in the dark for 10 min for colour development. The reaction was terminated by adding 50 µL of sulfuric acid stop solution to each well. The absorbance of each well was measured at 450 nm using a microplate reader within 15 min after termination, and the concentration of each target enzyme was calculated based on the standard curve.

Sequencing of 16S rRNA of fermentation substrates after pretreatment

The fermentation substrates subjected to two different pretreatment methods were analyzed by 16S rRNA gene sequencing. Total genomic DNA was extracted using a fecal DNA extraction kit (D3141, Guangzhou Meiji Biotechnology Co., Ltd., China). The V3-V4 hypervariable region of the 16S rRNA gene was amplified with specific

primers 341F/806R [29], and the resulting amplicons were used to construct Illumina sequencing libraries. Sequencing was performed on the Novaseq 6000 platform with a PE250 strategy. Raw sequencing data were quality-controlled and assembled to generate Clean Tags. These tags were then clustered into operational taxonomic units (OTUs) at 97% similarity using Usearch, with chimeric sequences removed during the process. Based on the OTU data, alpha diversity (Shannon and Simpson indices) and beta diversity (principal coordinate analysis based on Bray–Curtis distance) were analyzed with QIIME2 [6]. Taxonomic annotation of representative OTU sequences was carried out using the Silva database [23], enabling comparative analysis of microbial community structure and abundance across different pretreatment groups. The raw sequence data were stored under the registration number PRJNA1347783.

Metagenomic sequencing of solid residues after anaerobic fermentation

Metagenomic sequencing was performed on solid residues obtained after anaerobic fermentation from different treatment groups. Total DNA was extracted using a fecal DNA extraction kit (D3141, Guangzhou Meiji Biotechnology Co., China), and Illumina sequencing libraries were constructed and sequenced on the Novaseq 6000 platform with PE150 strategy [46]. After quality control of the raw data, Clean reads were obtained and assembled into contigs using MEGAHIT [17]. Protein-coding genes were predicted by MetaGeneMark, and a non-redundant gene catalog was constructed with CD-HIT [22]. Gene abundance was quantified using Bowtie and PathoScope [9]. Functional annotation was performed by aligning gene sequences against the KEGG, CAZy, and CARD databases with DIAMOND [32] to identify metabolic pathways, carbohydrate-active enzymes, and antibiotic resistance genes, respectively. Taxonomic annotation was carried out by comparing sequences to the Nr database using Kaiju [15]. Microbial diversity was assessed through alpha and beta diversity analyses using Scikit-bio and Vegan packages. Visualization of taxonomic and functional profiles was generated with ggplot2 and related tools. Statistical analysis was conducted in SPSS, employing one-way ANOVA followed by LSD post hoc testing, with a significance threshold set at $P<0.05$.

Statistical analysis

Data organization was performed using Microsoft Excel, and statistical analyses were conducted with SPSS 24.0. One-way analysis of variance (ANOVA) was employed to assess differences among groups for quantitative outcomes, with Tukey's test applied for post hoc multiple comparisons. Prior to performing the ANOVA, the assumptions of normality of residuals (assessed via the Shapiro–Wilk test) and homogeneity of variances (assessed via Levene's test) were verified, and both assumptions were met. A significance threshold of $P<0.05$ was used to determine statistical significance. Correlations between variables were examined using Spearman's rank correlation analysis in SPSS [21], with significance also defined as $P<0.05$. All graphical representations were generated using GraphPad Prism 8, TBtools, and R (version 4.1.3).

Results and discussion

Short-term aerobic pretreatment significantly enhances SCFAs production and optimizes product composition

This study systematically compared the effects of short-term aerobic pretreatment (SH) and conventional low-temperature thermal pretreatment (CK) on SCFAs production during dry anaerobic fermentation. The results showed that from day 3 of fermentation onward, SCFAs production in the SH group consistently exceeded that of the CK group, with statistically significant differences observed at multiple time points ($P<0.05$) (Fig. 1A). Over the entire fermentation period, the average total SCFAs yield in the SH group reached 148.01 ± 11.85 g, representing a significant increase of 21.31% ± 4.23% ($P<0.05$) compared to the CK group (116.13 ± 3.27 g) (Fig. 1A). In terms of product distribution, the SH group exhibited higher accumulation of acetate and propionate, whereas the CK group showed a relative preference for butyrate and isovalerate production (Fig. 1B). This divergence carries important implications for metabolic pathways: acetate and propionate are primary products of hydrolytic acidogenesis, and their accumulation generally reflects a more efficient and thorough hydrolysis process [2, 3]. In contrast, the formation of butyrate and isovalerate is often associated with specific microbial communities and distinct acidogenic phases, with isovalerate frequently derived from branched-chain amino acid degradation [34]. Thus, SH pretreatment likely facilitated more comprehensive organic matter solubilization, redirecting metabolic flux toward direct acid production rather than complex branched-chain transformations.

Evolution of physicochemical properties reveals the underlying mechanism of pretreatment strategy

The dynamic changes in key physicochemical properties of fermentation substrates under different pretreatment methods are shown in Fig. 2. For pH (Fig. 2A), the SH group

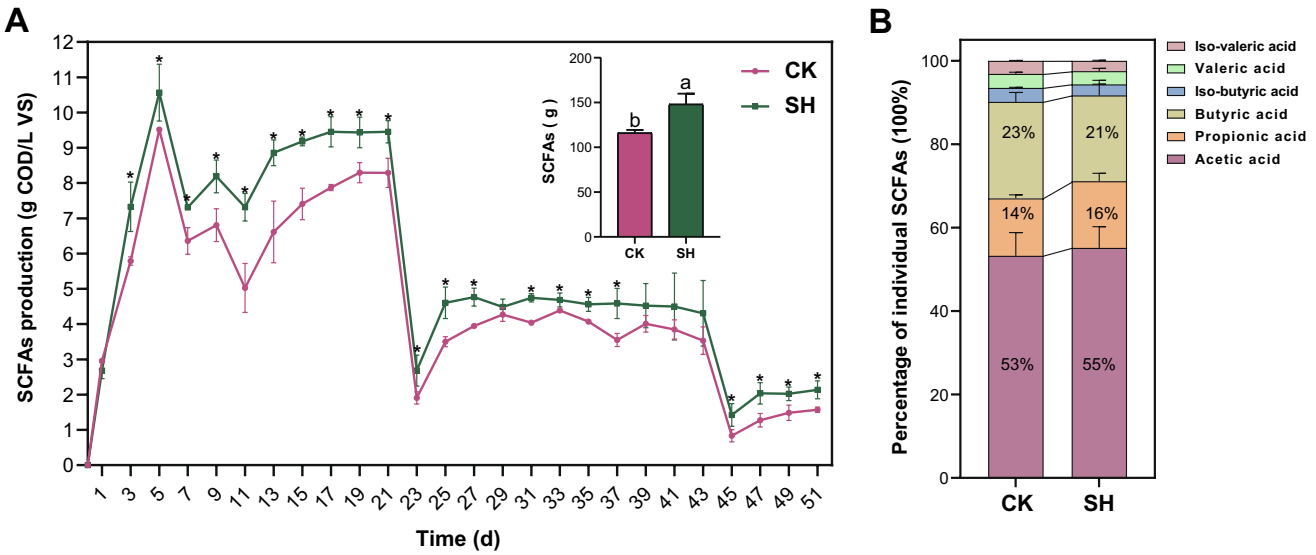


Fig. 1 SCFAs production and composition. **A** Dynamic profiles of SCFAs production during fermentation; **B** Relative composition of SCFAs components. Asterisks (*) and different lowercase letters indicate significant differences between treatment groups (P<0.05), those without marked Asterisks (*) have no significant difference

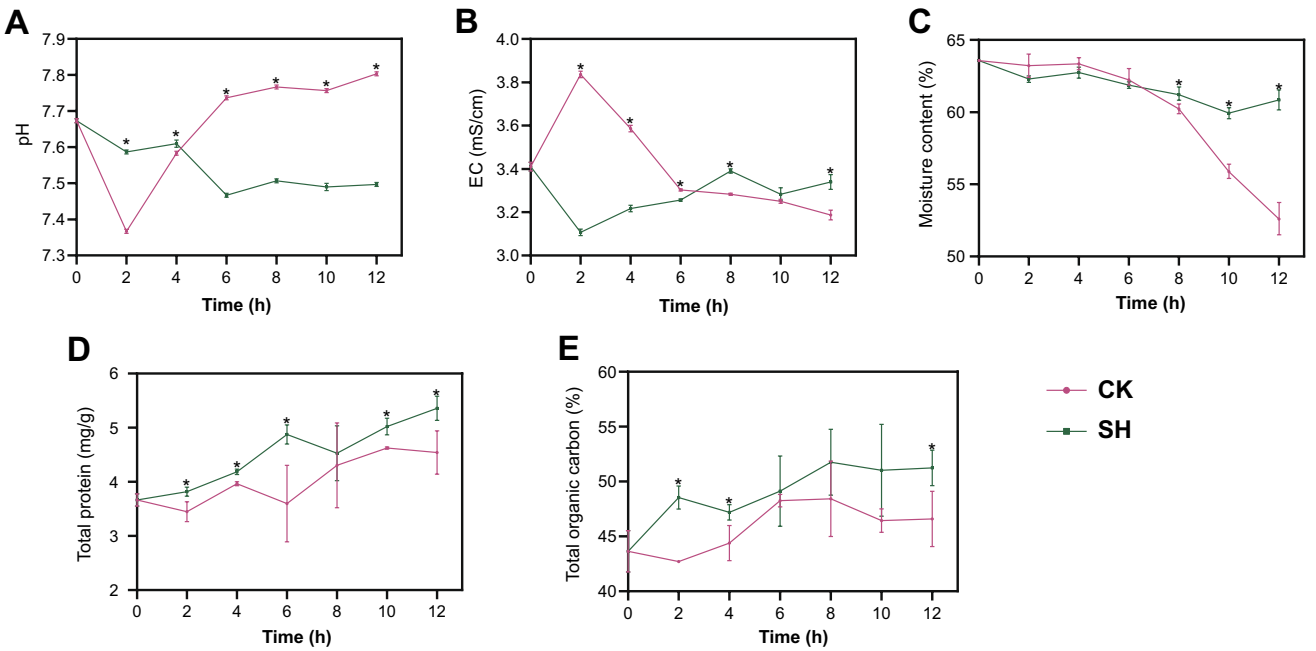


Fig. 2 Dynamic changes in physicochemical properties of the fermentation substrate during pretreatment. **A** pH; **B** Electrical conductivity; **C** Moisture content; **D** Total protein concentration; **E** Total organic carbon content. Asterisks (*) indicate significant differences between treatment groups at the same time point (P<0.05), those without marked Asterisks (*) have no significant difference

was significantly higher than CK at 2 and 4 h ($P<0.05$), whereas the CK group showed significantly higher values at 6, 8, 10, and 12 h ($P<0.05$). The final pH in the SH group (7.49±0.05) was significantly lower than that in CK (7.80±0.07) ($P<0.05$). As for electrical conductivity (Fig. 2B), the SH group exhibited significantly higher values than CK at 8 and 12 h, while the opposite was observed at 2, 4, and 6 h ($P<0.05$). The final conductivity in SH (3.34±0.04 mS/cm) was significantly higher than in CK (3.19±0.05 mS/cm) ($P<0.05$). In terms of moisture content (Fig. 2C), SH was higher than CK at 8, 10, and 12 h, whereas CK showed higher values at earlier time points (2, 4, and 6 h). The final moisture content in SH (60.84%±0.06%) was higher than that in CK (52.60%±0.03%). Total protein content (Fig. 2D) was consistently higher in SH across all time points, with significant differences observed at 2,

4, 6, 10, and 12 h ($P<0.05$). The final protein content in SH (0.936±0.021 mg/g) was significantly higher than in CK (0.854±0.037 mg/g) ($P<0.05$). Total organic carbon (Fig. 2E) remained higher in SH throughout the process, with significant differences at 2, 4, and 12 h ($P<0.05$). The final TOC in SH (52.25%±0.15%) was significantly higher than in CK (46.58%±0.09%) ($P<0.05$).

The continuous monitoring of these physicochemical indicators provides a theoretical basis for interpreting the differences in acid production under different pretreatments. The SH group ultimately exhibited lower pH, along with higher conductivity, moisture content, total protein, and total organic carbon (Fig. 2). These characteristics collectively established a micro-environment with high microbial activity and substrate load. The slightly lower pH (7.5 vs. 7.8) tends to moderately favour the growth and metabolism of hydrolytic and acidogenic bacteria [13, 51], although the effect is expected to be subtle and not a dominant driver under our experimental conditions. Elevated conductivity reflects increased ionic concentration in the system, likely associated with small organic acids and inorganic ions released during organic matter decomposition [26, 41]. The higher moisture content in SH improved mass transfer efficiency, promoting microbe-substrate interactions [43, 38]. Meanwhile, the significant increases in total protein and total organic carbon further confirm that SH pretreatment more effectively releases and retains organic substances, thereby supplying a more abundant substrate pool for subsequent fermentation.

Enhanced key enzyme activities demonstrate intensified metabolic pathways

Changes in the activities of key enzymes during subsequent anaerobic fermentation of substrates from different pretreatment methods are shown in Fig. 3. For α-glucosidase activity (Fig. 3A), the SH group exhibited higher values than CK at all time points, with significant differences at 2, 6, and 8 h ($P<0.05$). At the endpoint, the enzyme activity in SH (1.92±0.50 ng/g) was higher than in CK (1.56±0.12 ng/g), though not significantly. Protease activity (Fig. 3B) was significantly higher in SH at 2, 4, 6, 10, and 12 h ($P<0.05$), except for a slight (non-significant) decrease at 8 h. Final protease activity in SH (1.84±0.0.3 μg/g) was significantly higher than in CK (1.52±0.02 μg/g) ($P<0.05$). Acetate kinase activity (Fig. 3C) was significantly elevated in SH

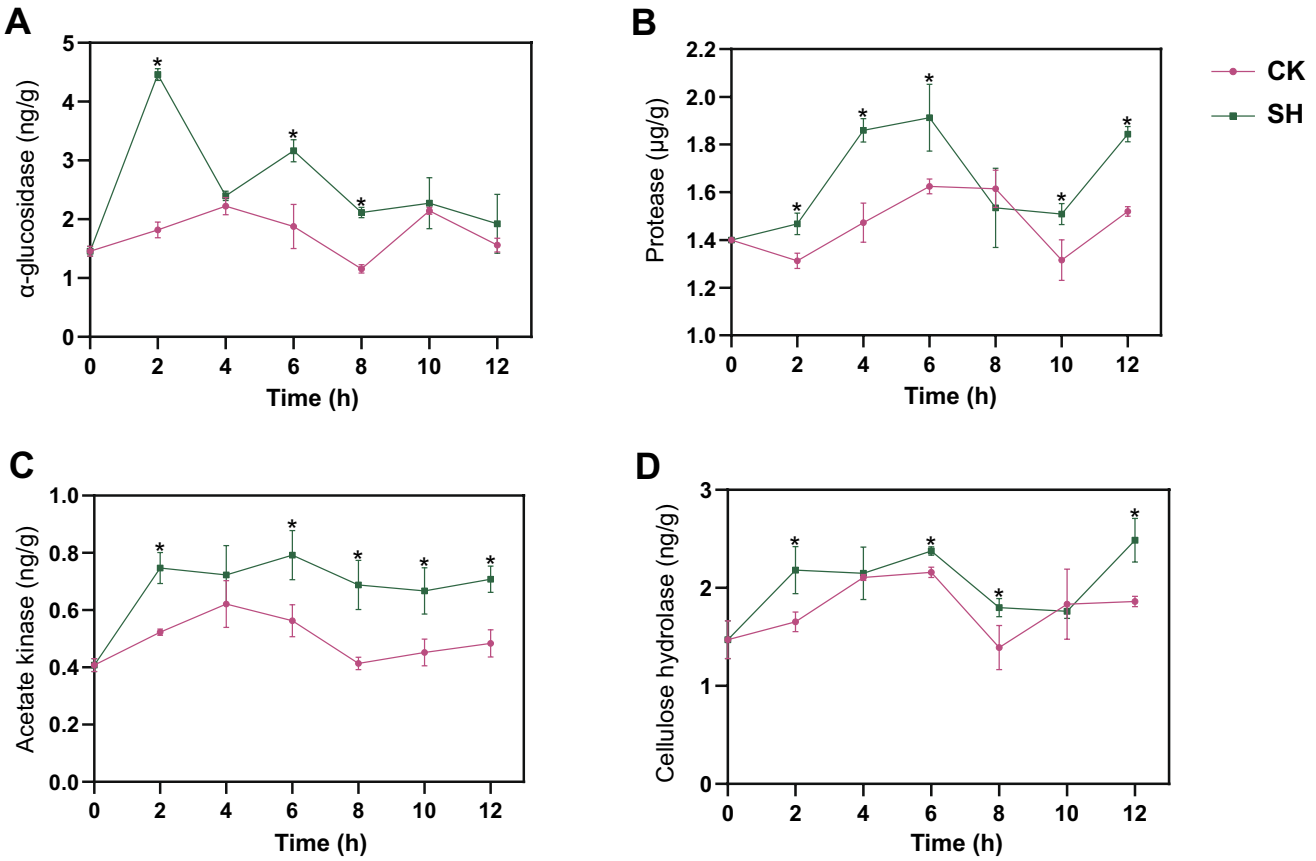


Fig. 3 Changes in key enzyme activities during pretreatment. **A** α-glucosidase; **B** Protease; **C** Acetate kinase; **D** Cellulose hydrolase. Asterisks (*) indicate significant differences between treatment groups at the same time point (P<0.05), those without marked Asterisks (*) have no significant difference

at 2, 6, 8, 10, and 12 h ($P<0.05$), with a non-significant decrease at 4 h. The final activity in SH (0.71 ± 0.05 ng/g) was significantly higher than in CK (0.48 ± 0.05 ng/g) ($P<0.05$). Cellulase activity (Fig. 3D) was significantly higher in SH at 2, 6, 8, and 12 h ($P<0.05$), with a non-significant reduction at 10 h. The final cellulase activity in SH (2.49 ± 0.22 ng/g) was significantly higher than in CK (1.86 ± 0.05 ng/g) ($P<0.05$).

From a biochemical perspective, enzyme activities directly reflect metabolic functions [7]. The results demonstrate that the SH group exhibited comprehensive advantages in the activities of multiple key hydrolases (α-glucosidase, protease, cellulase) and the acid-forming key enzyme (acetate kinase) (Fig. 3), indicating that the overall metabolic pathway—from the degradation of polysaccharides, proteins, and cellulose to acetate synthesis—was systematically enhanced in the SH system. Notably, the significant increases in cellulase and protease activities suggest that SH pretreatment may effectively disrupts lignocellulosic structures and protein macromolecules, which may be attributed to the abundant extracellular enzymes secreted by thermophilic and facultative aerobic microorganisms (such as *Bacillus* and *Streptomyces*) during the aerobic pretreatment process [25, 31]. Moreover, the elevated acetate kinase activity, a key enzyme catalyzing the conversion of acetyl-CoA to acetate, corresponded with the higher acetate yield in the SH group, confirming the enhancement of terminal acidogenic metabolism at the enzymatic level. This finding directly links macroscopic acid production performance with microscopic enzymatic catalysis, deepening the understanding of the pretreatment mechanism.

Restructuring of functional microbial community drives performance improvement

Microbial community structure is a key determinant of fermentation system functionality. Alpha diversity analysis revealed that the Shannon index of the SH group was significantly higher than that of CK ($P<0.05$) (Fig. 4A), indicating that short-term aerobic pretreatment effectively enhanced microbial diversity. Beta diversity analysis further demonstrated significant structural differences in microbial communities between the two groups (Fig. 4B). Analysis of high-abundance taxa at the genus level (Fig. 4C) showed that among the top 20 dominant genera, nine were significantly more abundant in SH than in CK ($P<0.05$), including *Paenibacillus*, *Lysinibacillus*, *Bacillus*, *Clostridium*, *Psychrobacillus*, *Sporosarcina*, *Planifilum*, *Streptococcus*, and *Streptomyces*. Most of these genera have been reported as typical hydrolytic and acidogenic functional bacteria. For instance, *Clostridium* and *Bacillus* secrete extracellular enzymes to decompose complex organic matter [5, 12], while *Paenibacillus* and *Streptomyces* are closely associated with cellulose and protein degradation [10].

Further correlation analysis indicated that *Paenibacillus*, *Lysinibacillus*, *Clostridium*, *Planifilum*, *Ureibacillus*, *Geobacillus*, and *Streptomyces* were significantly positively correlated with the activities of multiple key hydrolases, and these genera were generally more abundant in the SH group. Together, these findings form a coherent metabolic cascade: SH pretreatment selectively enriches specific functional microorganisms, thereby enhancing the activities of key hydrolytic and acidogenic enzymes, which in turn promotes organic matter degradation and ultimately increases SCFAs accumulation. This aligns with the concept of "function-driven community assembly" in microbial ecology [44], emphasizing that the goal of pretreatment extends beyond altering taxonomic composition to deliberately shaping a synergistic multi-microbial system with enhanced metabolic capabilities.

Environmental benefits: significant reduction in antibiotic resistance gene dissemination risk

Against the backdrop of increasing concern over the dissemination and accumulation of antibiotic resistance genes (ARGs) in the environment [55], assessing the environmental safety of fermentation products has become particularly important. This study analyzed the composition and distribution of ARGs in solid residues derived from two pretreatment methods—short-term aerobic pretreatment (SH) and conventional low-temperature thermal pretreatment (CK) (Fig. 5). The results showed that the types of ARGs present in the solid residues after 51 days of anaerobic fermentation were generally similar between the two groups (Fig. 5A), with fluoroquinolone, tetracycline, and macrolide resistance genes being the most dominant categories.

Further subgroup analysis of these three major ARGs classes revealed that the relative abundance of several high-risk ARGs was significantly lower in the SH group ($P<0.05$) (Fig. 5B), including *ermC* (fluoroquinolone), *tet(M)*, *tet(O)*, *tet(X)* (tetracycline), and *qnrB* (macrolide). These findings indicate that the short-term aerobic pretreatment offers superior environmental friendliness in mitigating ARG dissemination risks. The underlying mechanisms may involve two aspects: first, the thermophilic phase during aerobic pretreatment could directly inactivate certain potential pathogens harboring ARGs [1] (such as *Escherichia-Shigella*, which was significantly enriched in the CK group); second, the aerobic environment fostered a more competitive microbial community structure, potentially suppressing the proliferation of ARGs host bacteria through niche competition [43]. This phenomenon aligns with the principle of "competitive exclusion," whereby a functionally stable and

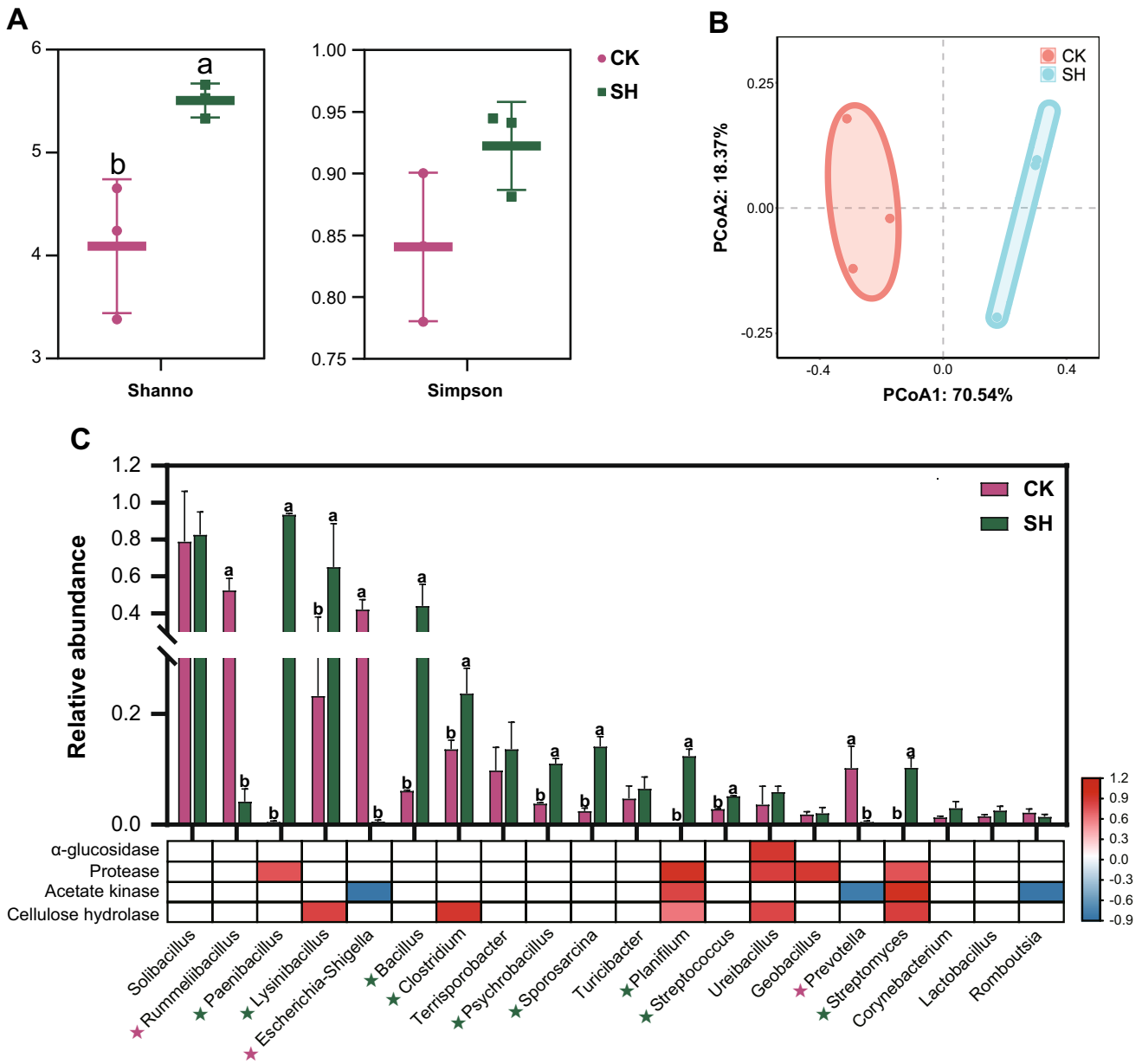


Fig. 4 Microbial community structure of fermentation substrates after pretreatment. **A** Alpha diversity indices; **B** Principal coordinate analysis of beta diversity based on Bray–Curtis distance; **C** Relative abundance of the top 20 bacterial genera. The heatmap shows correlations between genera and key enzyme activities (red indicates positive correlation, blue indicates negative correlation). Genera marked with asterisks (★) show significant differences between groups. Different lowercase letters indicate significant differences between treatment groups ($P<0.05$), those without marked letters have no significant difference

diverse microbial community can effectively resist colonization by exogenous pathogens and resistant bacteria [11]. In summary, while enhancing product value, the SH pretreatment also yields fermentation residues with lower environmental risks, providing an important guarantee for their subsequent resource utilization as organic fertilizers.

Economic feasibility: breakthrough advantages in energy consumption and cost control

Economic feasibility is a critical determinant for the practical adoption of any technology. To systematically evaluate the potential of short-term aerobic pretreatment (SH), this study conducted a comprehensive analysis of its energy consumption and economic performance (The specific accounting process is detailed in the Supplementary Materials Text A.2). The results showed that the average

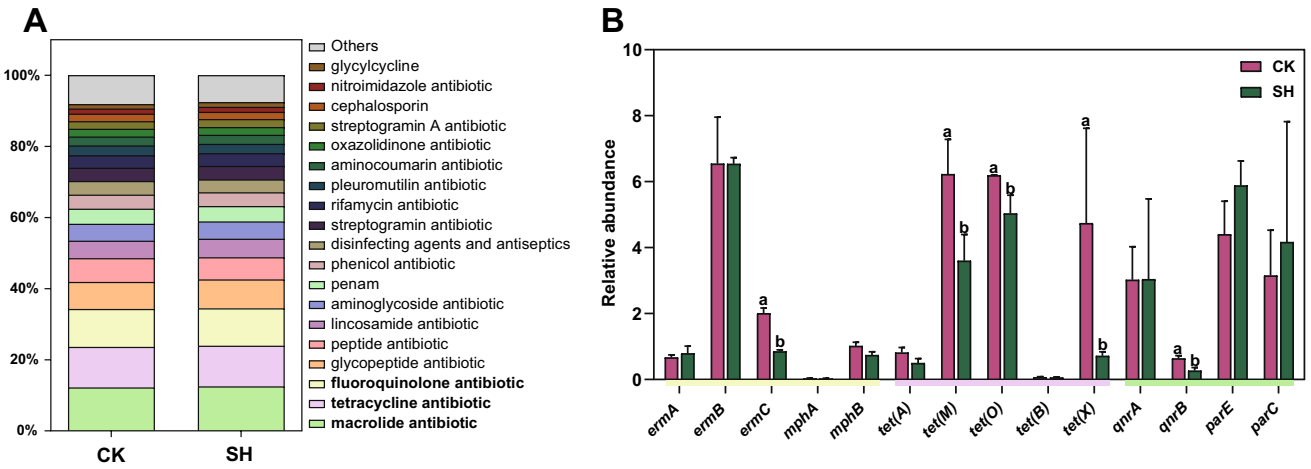


Fig. 5 Antibiotic resistance genes (ARGs) in fermentation residues from different pretreatments. **A** Composition of ARG types; **B** Relative abundance of key ARGs. Different lowercase letters indicate significant differences between treatment groups ($P<0.05$), those without marked letters have no significant difference

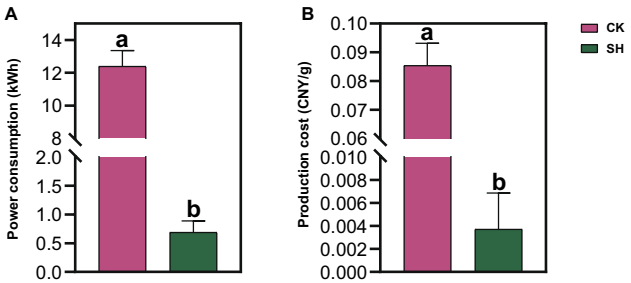


Fig. 6 Cost analysis of different pretreatment methods. **A** Energy consumption; **B** Pretreatment cost per gram of short-chain fatty acids produced. Production cost (CNY/g) = Electricity consumption × Electricity price ÷ Total SCFAs production. For conventional low-temperature thermal pretreatment, energy consumption was mainly from oven heating; for short-term aerobic pretreatment, energy consumption was primarily from aeration. Different lowercase letters indicate significant differences between treatment groups ($P<0.05$)

electricity consumption of SH was 0.69 ± 0.20 kWh, significantly lower than the 12.4 ± 0.95 kWh ($P<0.05$) required by conventional low-temperature thermal pretreatment (CK) (Fig. 6A), representing a 94.48% ± 1.48% reduction in energy use. Further cost analysis revealed that the pretreatment cost per gram of SCFAs produced was only ¥0.0037 for SH, considerably lower than the ¥0.0854 ($P<0.05$) for CK (Fig. 6B), equivalent to merely 4.37% ± 0.15% of the conventional pretreatment cost. These findings demonstrate that SH effectively overcomes the high energy consumption and cost barriers associated with traditional thermal methods, positioning it as a promising pretreatment strategy with high energy efficiency and economic viability.

The notable advantages of SH stem from its reliance on microbial metabolic heat for self-sustaining temperature elevation, requiring minimal external energy input, whereas conventional thermal pretreatment depends heavily on continuous electricity consumption for heating. As a result, SH not only significantly reduces operational costs but also aligns closely with global carbon neutrality goals, highlighting its strong potential for broad implementation. This technology is expected to shift organic solid waste treatment from a conventional energy-intensive model toward an energy-self-sufficient or even energy-producing paradigm, providing a key pathway for establishing a sustainable waste valorization system.

Conclusion

This study demonstrates that short-term aerobic pretreatment enhances the solubilization and hydrolysis of organic matter, thereby establishing a robust substrate foundation for efficient SCFAs synthesis. Furthermore, by selectively enriching functional microbial taxa such as *Clostridium*, *Bacillus*, *Paenibacillus*, and *Streptomyces*, the pretreatment significantly strengthens the metabolic activity of key hydrolytic and acidogenic enzymes, ultimately increasing total SCFAs production by approximately 21.31%. Beyond these bioenhancement effects, the strategy exhibits outstanding comprehensive advantages: it reduces energy consumption during pretreatment by roughly 94.48%, cuts the unit acid production cost to only about 4.37% of that of conventional thermal pretreatment, and substantially lowers the environmental risk of antibiotic resistance gene dissemination. This approach thus provides a viable and sustainable technological pathway for high-value, low-carbon valorization of organic solid waste.

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Author contributions Bangjie Xiong: Experiment design, conducted experiments, data analysis and manuscript writing. Ju Chen:Conducted experiments. Liangbin zhao: Conducted experiments. Mengting Chen: Conducted experiments. Xiaoxia Hao: Manuscript review. Weiping Sang: Conducted experiments. Zhenzhen Guo: Conducted experiments. Dongmei Jiang: Manuscript review. Shuo Liu: Data analysis. Li Zhu: Project administration and funding acquisition. Lin Bai: Experiment design, manuscript review, editing and funding acquisition.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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